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Meta-Analysis of Emotion Recognition Intervention Effects and Influencing Factors in Autism Spectrum Disorder

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Abstract: *Emotion recognition has been identified as critical in the development of social communication and interaction with individuals having autism spectrum disorder (ASD). However, little is known about the effects of the intervention and moderator factors with quantitative research. To carry out an analysis of effect size on the intervention emotion recognition of ASD, and to propose an optimal intervention plan, a meta-analysis was conducted across 22 randomized controlled trial studies involving 817 individuals with ASD, with the aim of analyzing the effectiveness of moderator variables. The findings of this study showed that the effect size of emotion recognition intervention for ASD reached a medium level (d = 0.69), and the effect size of the technology-based intervention (d = 0.90) was significantly greater than that of the non-technology-based intervention (d = 0.37). Though technology-based intervention may be a promising method on emotion recognition for ASD individuals, however, more well-controlled studies are needed.*

Autism spectrum disorder is a developmental condition that involves difficulties with social interaction and communication, and by restricted and repetitive behavior (Lord et al., 2020). According to the 2020 statistics of the Centers for Disease Control and Prevention (CDC), the prevalence rate of autistic children under 8 years old in the United States has reached 1/54 (Knopf, 2020). Understanding the emotional and mental state of others plays a major role in the social and communicative characteristics of ASD (Huang et al., 2017). Fundamental to this is the ability to recognize and discriminate emotional expressions, a skill which is present from at least 10 weeks of age in typically developing infants and continues to develop across childhood (Cao et al., 2017). Individuals with ASD have shown delays in the development of this ability. These delays are demonstrated through tasks assessing emotion recognition from facial expressions, vocal intonation, and body language (Fridenson-Hayo et al., 2017). Meanwhile, in ecological life-like tasks, require the integration of emotional cues from different perceptual channels in context (Sprung et al., 2015). Emotion recognition has the ability to modulate arousal and emotional responses in the service of adaptive and socially appropriate behaviors (Bal et al., 2010). However, 50%-85% of individuals with ASD possess clinically significant emotion recognition difficulties (Bora & Pantelis, 2016).

These emotion recognition difficulties are associated with altered attentional, perceptual, cognitive, and neural processes (Golan et al., 2010). Individuals with ASD process faces differently and show reduced attention to faces and facial expressions. This may be due to the mentalistic and emotional information conveyed by the eyes and facial expressions, which may be hard to read for individuals with ASD (Lozier et al., 2014). However, there is evidence of reduced attention to faces by looking at the eyes in particular among children with ASD (Trevisan & Birmingham, 2016). This reduced interest in faces is likely to impair their face processing skills so that children with ASD do not become 'face experts' like their typically developing peers (Weiss et al., 2018a). Also, it has been suggested that children with ASD focus on the mouth area during social exchanges because they may be trying to obtain verbal information, in the absence of attending to the eyes, as an alternative 'scaffold' with which to interpret social interactions (Sprung et al., 2015).

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Previous Meta-Analysis

Although previous researches conducted on emotion recognition studies in ASD is informative, nevertheless, most of the studies only focused on the impairments in emotion recognition and lack of evidence on the intervention programs. For example, two previous meta-analyzes found that participants with ASD scored significantly lower than the matched neurotypical controls on emotion recognition tasks presenting images of facial expressions and other nonverbal body cues or from tasks presenting facial expression stimuli only (Lozier et al., 2014; Uljarevic & Hamilton, 2013). One meta-analysis found that facial emotion recognition ability represents an important social cognitive ability given its relation to real-world social behavior and other characteristics and cognitive abilities (Schäfer et al., 2017). The moderator analysis, by Uljarevic and Hamilton (2013) found that neither age, intelligence, nor task factors (e.g., labeling vs. matching paradigms) moderated the strength of the overall effect size. On the contrary, Lozier et al. (2020) emphasized that age, but not intelligence, moderated the strength of the group differences between ASD and controls, such that the ASD deficits increased with age (Lozier et al., 2014). To sum up, there are still potential moderators in emotion recognition in ASD participants.

Present Studies

The existing studies are an extension of the work done based on the previous meta-analysis. The intervention programs of emotion recognition in ASD participants include cognitive behavior therapy, painting therapy, music therapy, and psychological counseling. Although these programs may develop emotion recognition in ASD children, the potential problems with these interventions still need to be addressed. For example, cognitive behavior theory intervention relies heavily on cognitive ability of children with ASD and may have limited intervention effects. Psychological counseling requires individuals with ASD to have certain speaking comprehension ability and emotional expression ability, but most of the individuals with ASD lack such ability, thus resulting in limited intervention effects. To date, technology-based interventions (TBI) are represented by computer-based intervention programs and virtual reality technology has shown unique advantages that are gradually applied to emotion recognition intervention of ASD participants (Grynszpan et al., 2014). Firstly, the TBI cater to the characteristics of the visual advantages in individuals with ASD, and the multi-sensory stimulation greatly enhances the interest and concentration of the emotion recognition. Secondly, TBI is easily acceptable by the parents and intervention teachers. Thirdly, TBI can reduce labor costs thus saving intervention and training costs. Lastly, TBI can avoid the interference of subjective factors and the integrity and objectivity of intervention effects.

Despite the significant number of emotion recognition studies in ASD, little is known on the intervention effects among various intervention programs. Given the previous studies, it is important to establish the effectiveness of interventions aimed at improving emotion recognition of individuals with ASD and to assess whether there are any specific intervention characteristics that are associated with more favorable outcomes. The aim of this meta-analysis was to evaluate the overall efficacy of emotion recognition. Additionally, it examines and analyses the influence of different intervention characteristics: the number of participants, the length of intervention period, the intensity of intervention, environment setting, parents involvement, intervention sites (country), and intervention methods. Notably, the RCT study design was selected because it is often considered the golden standard for clinic intervention, and it can avoid the selectivity bias and the interference of unrelated factors (Savitz, 2003).

Method

Search. This study follows the PRISMA guidelines (which stands for Preferred Reporting Items for Systematic Reviews and Meta-Analysis) to aid in the transparent reporting of systematic reviews and meta-analysis. Following PRISMA guidelines, Figure 1 shows a flow diagram of studies to include when reporting on the meta-analysis. Three authors who did not report sufficient data to calculate the effect sizes in their published abstracts were contacted, as a result, relevant data were

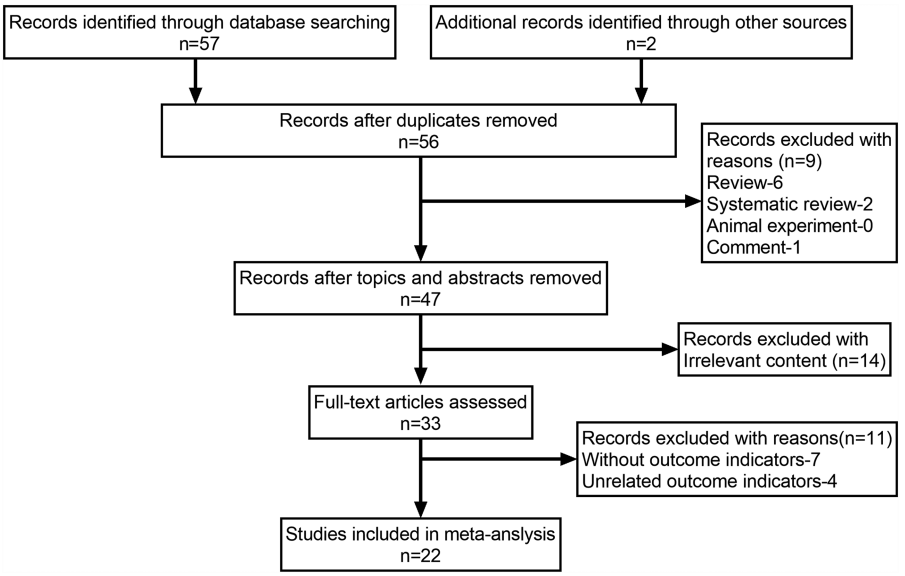


Figure 1. Flow diagram of the study selection process.

received from two of them. Furthermore, we contacted the authors of two master's theses (Yu Hong Yan & Juan Wang), even though they were not published in the database.

There were six databases (Web of Science, EBSCO, ERIC, SAGE, CNKI, WAN FANG). Subject words, random words, and Boolean logic operation were used in the retrieval. The retrieval time was from the establishment of the database to September 10, 2020. Retrieval type: (Autism OR ASD OR Autism Spectrum Disorder OR Developmental Disorder) AND (Emotion Recognition OR Emotion Competence OR Emotion Expression OR Affection Recognition OR Affection Ability OR Affection Competence) AND (RCT OR Randomized Controlled Trials OR Randomized Controlled Trial). Additionally, an extensive manual review of reference lists of relevant studies and review articles extracted from the database searches was conducted. The studies that were determined to be related to the topic of emotion intervention or emotion recognition intervention were selected for further examination.

Selection criteria. Studies included in the meta-analysis must comply with the following criteria: (1) Literature published in English and Chinese; (2) ASD diagnosis must be established through a reliable measure; (3) Studies must report on a randomized controlled trial; (4) The research purpose should include ASD emotion recognition or emotion recognition ability and relevant variable data; (5) The experimental group only received emotion recognition intervention, while the control group received conventional intervention or no intervention; (6) Provide sufficient data to perform effect size analyzes (i.e., means and standard deviations, *F*-values, or change scores), If all criteria (6) were met, except for criterion, the authors were then contacted for additional data.

Data abstraction. Two researchers (first and third author) independently extracted data according to the inclusion criteria and performed an inter-rater agreement analysis using Kappa, and any disagreement arising were resolved by discussion. Additionally, continuous and categorical variables were extracted for the purpose of conducting moderator analysis.

Methodological quality. The Cochrane risk of bias assessment tool was used (Figure 2 and Figure 3). Evaluation included random assignment methods, allocation scheme concealment, blinding of implementer and participants, integrity of

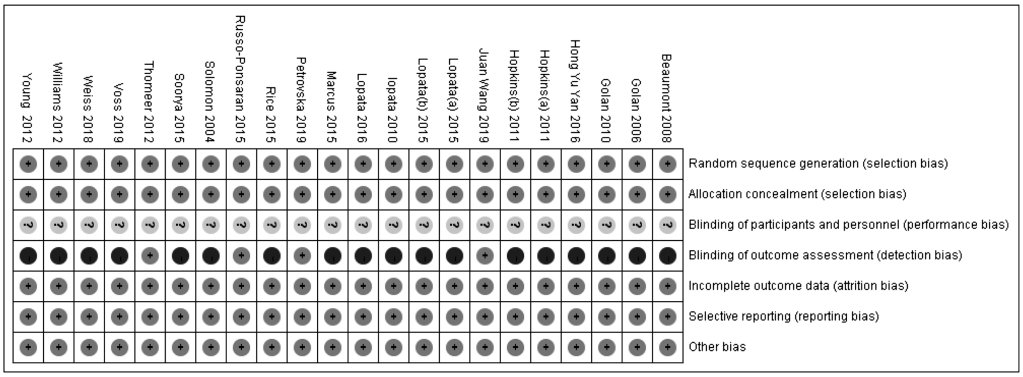


Figure 2. Methodological quality of included studies. ***Note.*** **"+" indicates reach standard; "indicates cannot reach standard;?" means unclear contents.**

outcome data, selective reporting of findings, and bias from other sources. To ensure the reliability of the screening process and avoid subjective bias, a methodological quality evaluation was conducted by two independent raters (first and third author). Any disagreements arising were resolved by discussion. According to the Cochrane evaluation criteria, the quality criteria for 4 or more items were fully met thus indicating the low-risk bias. Likewise, the quality criteria of 2 or more items were also met, indicating the medium-risk bias. If only one item meets the quality criteria, indicating the high-risk bias.

Quantitative Data Synthesis

Calculation of effect sizes. CMA2.0 software was used for analysis. In CMA2.0, the sample size, post-test mean, and standard deviation of the experimental group and the control group were regarded as input, and the effect sizes were calculated. The effect sizes of emotion recognition used Cohen's *d* and its 95% confidence interval. The magnitude of Cohen's *d* corresponds with Cohen's interpreting effect sizes as small (0.2), medium (0.5), and large (0.8).

A random-effects model was used in the meta-analysis due to the fact that the studies in the meta-analysis do not share a common true effect. Furthermore, the existence of significance between the studies is also assumed. As the intervention effect of emotion recognition in ASD may be affected by the moderator variables (i.e., the length of intervention period, the intensity of intervention, and different intervention methods), therefore, the random-effects model was adopted. In the

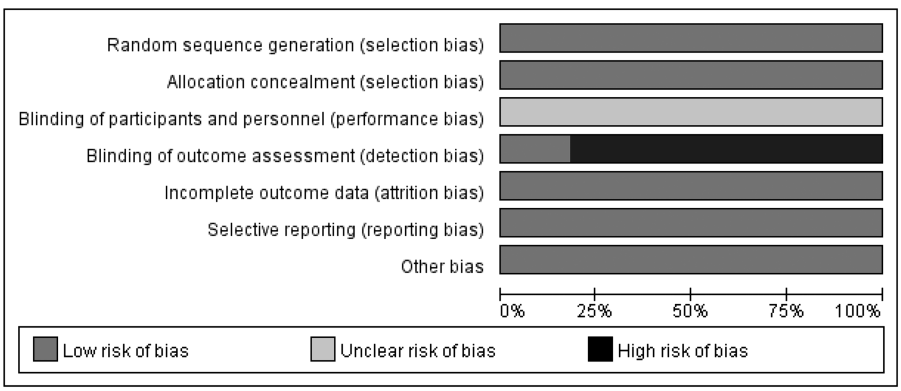


Figure 3. Distribution of the methodological quality of included studies.

following meta-analysis, a heterogeneity test will be further used to verify the model selection.

Publication bias. For the risk publication bias, the fail-safe N was calculated to estimate the number of unpublished studies, with an effect size of zero, to nullify the significant effect. The effect size can be considered robust if the required number of studies to reduce the overall effect size to a non-significant level exceeds $5k+10$, where k is the number of studies included in the analysis. Additionally, the risk of publication bias was also assessed using the Egger's test, and the linear regression method to test. Generally, the intercept of the linear regression equation and its 95% confidence interval is first obtained, and then the hypothesis test is conducted to ascertain whether the intercept is 0. If it is not significant, it indicates that there is no publication bias. Additionally, we constructed a funnel plot to give a visual idea of potential publication bias. Following convention, the precision of the studies is plotted along the y-axis with more precise studies (e.g., larger N) at the top and less precise studies (e.g., smaller N) at the bottom of the graph. Studies were plotted along the X-axis depending on the estimates of the effect sizes. The asymmetrical distribution of the studies on the effect size estimates, around the mean effect size, is assumed with more variability at the bottom and the more precise studies at the top, which is closer to the estimated mean effect size. It is assumed that if there are more studies at the right (high effect sizes) than at the left (small or null effects) side of the graph, the presence of publication bias is suggested. Finally, the Duval and Tweedie's Trim and Fill method was applied which uses an iterative process in which the most extreme small studies are removed and a new effect size is computed until the funnel plot is symmetrical. Furthermore, the number of studies that might be missing on the left side of the funnel plot was examined in order to obtain the effect size including the estimated missing studies.

Moderator variables. There are two approaches to analyze the moderator variables. Firstly, the meta-regression analysis examines whether the effect sizes vary as a function of following continuous variables: the length of intervention period (in weeks), the intensity of intervention (the average time over a week). Secondly, Q statistics determine whether the effect sizes vary as a function of the following categorical moderators: environmental setting (home, school, clinic, and college campus), country (North America, Europe, and others), parents' involvement (Yes and No), method (TBI and non-TBI).

The authors grouped the studies according to the categorical variables being assessed. Then, the authors calculated the Q_{between} using the effect weights with pooled estimates to determine whether heterogeneity exists between the groups. If the Q_{between} value was significant ($p < 0.05$), the grouping variable was considered a moderator.

Results

Study selection. Figure. 1 illustrates the study selection process. Of the 59 papers initially identified, only 56 papers were selected after the duplication check. The remaining 47 articles were due to excluding 8 review articles and 1 comment article. The 33 papers were included after excluding irrelevant content (n = 14). Finally, the remaining 22 papers were included after excluding the unrelated outcome indicators (n = 11). The inter-rater agreement on the task classification was high with a Kappa of 0.87 ($p < 0.001$), *95%CI* [0.73, 1.03]. There was no inter-rater disagreement between the two independent raters.

Study characteristics. Table 1 provides a description of the characteristics of the studies included in the present analysis. The studies were conducted in six different countries (Canada, China, Australia, Macedonia, UK, and USA) and included a total of 817 participants. The majority of IQ assessment tools used was the WISC and half of the intervention studies involved parents (n = 11), and the studies were conducted in four different environment settings (home, school, clinic, and college campus). Most of the studies used TBI (n = 14). The age of participants ranged from

TABLE 1

Description of Studies

Studies	*Age range*	*IQ*	*N*	*Environmental setting*	*Parents involvement*	*Country*	*Method*	*Length of period*	*Intensity of intervention*
(Voss et al., 2019)	6-12 years	Stanford Binet	71	Home	No	US	Superpower Glass	6 weeks	20 min/day
(Weiss et al., 2018b)	8-12 years	WASI-II	68	School	Yes	Canada	CBT	14 weeks	20 min/day
(Lopata et al., 2017)	7-12 years	WISC-IV	36	Clinic	No	US	Psychosocial treatment	5 weeks	20 min/day
(Thomeer et al., 2015)	7-12 years	WISC-IV	43	School	Yes	US	Mind Reading	12 weeks	60 min/day
(Soorya et al., 2015)	8-11 years	N/A	69	School	No	US	Mind Reading	12 weeks	60 min/day
(Rice et al., 2015)	5-11 years	WISC-IV	31	School	No	US	Face Say	10 weeks	25 min/day
(Lopata et al., 2015)	7-12 years	WISC-IV	47	College campus	Yes	US	Psychosocial treatment	5 weeks	20 min/day
(Williams et al., 2012)	4–7 years	WPPSI-III	39	College campus	No	Australia	Transporters Animation	4 weeks	15 min/day
(Lopata et al., 2010)	7-12 years	WISC-IV	36	School	Yes	US	Social Training	5 weeks	20 min/day
(Beaumont & Sofronoff, 2008)	7-11 years	WISC-III	49	Clinic	Yes	Australia	Computer Games	6 weeks	45 min/day
(Hopkins et al., 2011)	6-10 years	KBIT	24	College campus	No	US	Face Say	6 weeks	25 min/day
(Hopkins et al., 2011)	6-10 years	KBIT	25	College campus	No	US	Face Say	12 weeks	25 min/day
(Petrovska & Trajkovski, 2019)	7-15 years	N/A	32	school	Yes	Macedonia	Computer Games	12 weeks	25 min/day
(Solomon et al., 2004)	8-12 years	WISC-III	10	College campus	Yes	US	Emotion Course	8 weeks	30 min/day
(Thomeer et al., 2012)	7-12 years	WISC-IV	35	College campus	Yes	US	Emotion Course	20 weeks	90 min/day
(Young & Posselt, 2012)	4-8 years	WISC-IV	25	Home	Yes	Australia	Transporters Animation	5 weeks	20 min/day
(Golan & Baron-Cohen, 2006)	N/A	WISC-IV	41	Home	No	UK	Mind Reading	3 weeks	45 min/day
(Golan et al., 2010)	4-7 years	WISC-IV	38	College campus	Yes	UK	Transporters Animation	10 weeks	60 min/day
(Russo-Ponsaran et al., 2016)	8-15 years	WISC-IV	25	College campus	No	US	MiX Game	4 weeks	45 min/day

(continued on next page)

TABLE 1 (Continued)

Studies	*Age range*	*IQ*	*N*	*Environmental setting*	*Parents involvement*	*Country*	*Method*	*Length of period*	*Intensity of intervention*
Juan Wang (2019)	4-8 years	WISC-IV	12	College campus	No	China	Transporters Animation	6 weeks	60 min/day
Hong Yu Yan (2016)	4-9 years	WISC-IV	14	College campus	No	China	Transporters Animation	6 weeks	40 min/day
(Lopata et al., 2016)	7-12 years	WISC-IV	47	Clinic	Yes	US	Psychosocial treatment	6 weeks	30 min/day

Note. WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence; KBIT, Kaufman Brief Intelligence Test; CBT, Cognitive Behavior Therapy.

4 to 15 years old. The length of intervention period ranged from 3 to 20 weeks (mean = 8.05, SD = 4.10), with the intensity of intervention between 15 to 90 min/day (mean = 30.36, SD = 19.08).

Methodological quality of included studies. All the literature achieved a low-risk bias. Among them, the quality score of four pieces of literature reached 6 points, which were high-quality literature. In general, the quality of the 22 studies that were included is high (see Figure 2 and Figure 3).

Meta-Analysis Results

Overall effect test. A heterogeneity test was performed on the 22 studies. Table 2 and Figure 4 show that $I^2 = 62.78\% > 50\%$, and Q Test $p < 0.1$, which suggested that heterogeneity in the meta-analysis, the random-effects model can be selected. Besides, the effect sizes of emotion recognition intervention in ASD ranged from 0.45 to 0.93, with an average effect size of $d = 0.69$. Therefore, the overall effect size of this study is medium.

Test of publication bias. In this study, a subjective judgment was made to determine whether publication bias existed. As shown in Figure 5, asymmetrical distribution was scattered on both sides of the total effect quantity, with more on the right side, indicating the possibility of publication bias. As shown in Table 3, the Fail-Safe N method for analysis continues to be used. When the alpha was at 0.05, 430 more articles were needed to overturn the existing conclusion, and, therefore, the results may be considered robust. According to Egger's test analysis, the intercept was 4.58 $p = 0.0005 < 0.05$, indicating publication bias in this study. Given the inconsistency of the three test results, it is necessary to adopt the Duval and Tweedie's Trim and Fill method to address the question concerning the impact of publication bias on observed effect size. Figure. 5 shows that there were seven missing studies in which the effect size is less than the average effect sizes, and the additional data would reduce the average effect size. After correction, it shows $d = 0.41$, *95% CI* [0.14, 0.68]. Based on the above

TABLE 2

Effect Size and Heterogeneity Test of ASD Emotion Recognition Intervention

Model	*N*	*d*	*95% CI*	*Heterogeneity Test*		
				df	I^2	*p*
Random-effects	*22*	*0.69*	*[0.45, 0.93]*	*21*	*62.78*	*<0.0001*

analysis, although the publication bias may exist in the meta-analysis, the conclusion is still valid.

Moderator variables results. The heterogeneity test found that there was heterogeneity in the effect sizes between studies, and there may be moderator variables. Whether its the environment setting, the parent's involvement, the intervention methods, or the country playing a moderator role was analyzed. It can be concluded from Table 4 that the effect size was significantly moderated by the intervention method ($Q_{between}$ = 5.92, p = 0.015 < 0.05): the effect size of the TBI (Z = 4.74, p < 0.001, d = 0.90) was significantly higher than that of the non-TBI (Z = 3.38, p = 0.001, d = 0.37). The intervention setting, the parent's involvement, and the country were independent of the heterogeneity of the effect size.

This study also analyzes two continuous variables, namely the length of intervention period and the intensity of intervention. Figure 6, Figure 7, and Table 5 show that the regression model results were not significant, and the length of intervention period and the intensity of intervention could not predict the results. In other words, with the increase in the length of intervention and intensity, the effect on the emotion recognition intervention was not significant.

Discussion

The meta-analysis conducted a quality assessment and publication bias analysis of the 22

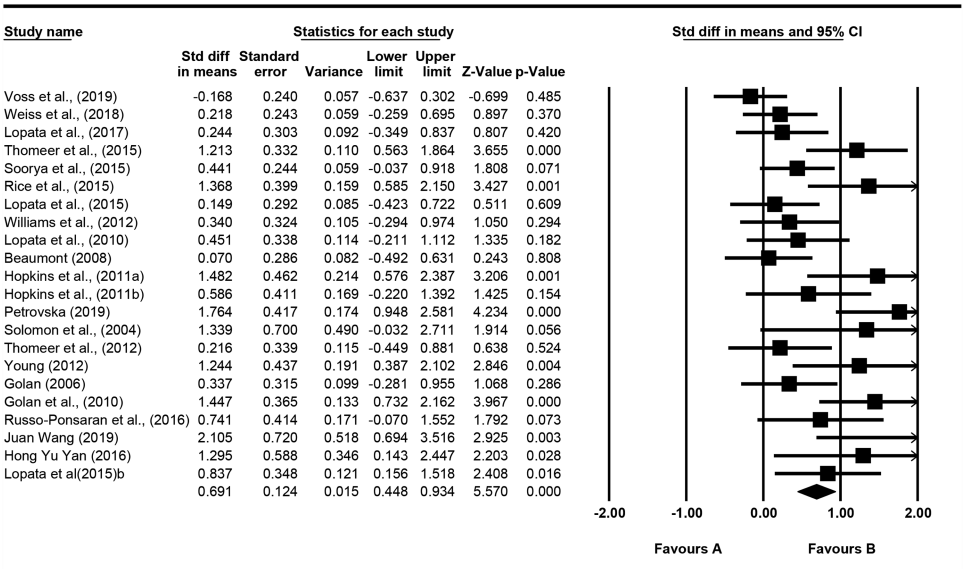


Figure 4. A forest plot for ASD emotion recognition intervention

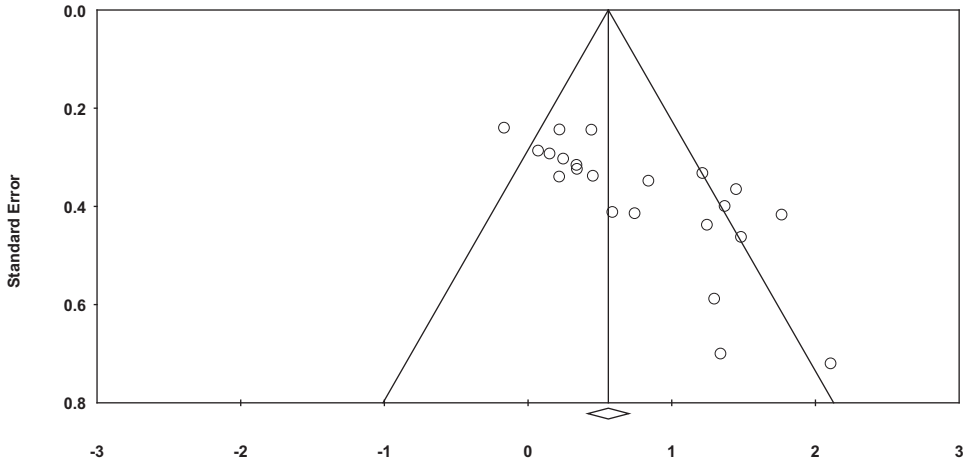


Figure 5. A funnel plot for ASD emotion recognition interventions.

included studies, reaching a significant medium level (d = 0.69). This conclusion is also consistent with the findings of Berggren et al. (2018). Thirteen eligible studies on training emotion recognition in ASD between the ages of 4 to 12 were identified which indicated that training may improve emotion recognition (Berggren et al., 2018). The findings of this study strongly suggest an overall positive effect of emotion recognition intervention for individuals with ASD. Additionally, this study provides robust data to demonstrate the positive effect of emotion recognition intervention on individuals with ASD and to improve the confidence of intervention teachers. As a fundamental part of social skills, emotion recognition plays a crucial role in the social interaction of individuals with ASD (Wieckowski & White, 2017). Therefore, this finding also provides necessary practical support for ASD participants and makes it possible for them to return to mainstream society.

TABLE 3

Test Results of Publication Bias

Nfs	*Egger's intercept*	*LL*	*UL*	*p*
430	*4.58*	*2.71*	*6.45*	*0.0005*

Note. Nfs: Fail-Safe N; *LL* and *UL* represent the lower and upper limits of the 95% confidence interval of the *Egger's*.

On the other hand, most of the current studies are mainly focused on intervention on the high-functioning autism, and it is debatable whether the findings can be extrapolated to all ASD spectrum. Although low-functioning autism may affect intervention effects because of difficulties in understanding commands and communicating, nevertheless, low-functioning autism should not be a reason to exclude interventions. Even if the findings of this study support the positive intervention effects, two potential issues should draw the researcher's attention. First, a large number of reports fail to conduct a longitudinal study. Some reviews have shown that emotion recognition ability in individuals with ASD may decrease as time passes (DeRosa et al., 2018; Karaminis et al., 2016; Yao et al., 2016). Therefore, we consider that the study on emotion recognition should emphasize on the longitudinal study to ensure lasting effectiveness. Second, attention should be paid to the difficulty in generalizing on emotion recognition. One study suggested that emotion recognition intervention can be carried out by combining the TBI with conventional teaching (Sharma et al., 2017). One interpretation of the study is that TBI is more powerful and attractive to children with ASD and is easy to intervene. However, the conventional social skill courses conducted in the form of group, emphasized the intervention in a real social

TABLE 4

Analysis of the Moderator Variables on Emotion Recognition Intervention

Moderator Variables	*Groups*	*k*	*d*	*LL*	*UL*	$Q_{between}$	*p*
Environmental setting	Clinic	3	0.35	−0.08	0.78	3.77	0.29
	College campus	10	0.83	0.44	1.21		
	Home	3	0.40	−0.34	1.14		
	School	6	0.85	0.37	1.32		
Parents involvement	No	11	0.64	0.30	0.99	0.15	0.70
	Yes	11	0.74	0.38	1.10		
Method	Non-TBI	8	0.37	0.16	0.58	5.92	0.02
	TBI	14	0.90	0.53	1.28		
Country	Europe	3	1.15	0.27	2.04	2.05	0.36
	North America	14	0.56	0.29	0.82		
	Others	5	0.84	0.18	1.50		

Note. *LL* and *UL* are the lower and upper limits of the 95% confidence interval of the effect size corresponding to the result variables; $Q_{between}$ indicates the *Q* test for inter-category heterogeneity.

situation. Therefore, the combination of these two intervention methods can solve the generalization problem of emotion recognition in individuals with ASD.

Analysis of Moderating Variables

The overall effect size for emotion recognition in ASD individuals was moderated by different intervention methods. The TBI method significantly improved intervention effects when compared to the non-TBI. There are three possible reasons to interpret results. First, the TBI approach emphasizes the diversity of sensory stimuli and the high efficiency of the intervention process (Grynszpan et al., 2014). Stimulation develops from the initial visual material to audio material and then to audiovisual material, which changes the process from static to dynamic stimulation. Also, TBI can enrich the sensory experience of individuals with ASD (Jaliaawala & Khan, 2020). For example, "Avatars" and "Mix" are designed in emotion recognition intervention software and they can provide individuals with ASD with help and learning strategies, such as instructing ASD individuals to pay attention to minor changes in the eye area and to achieve good training results. Second, TBI can present dynamic and personalized

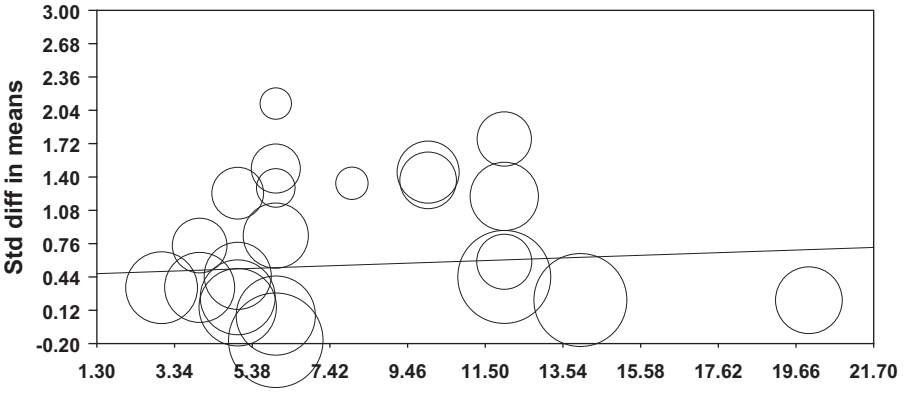


Figure 6. A meta-regression of the length of intervention period.

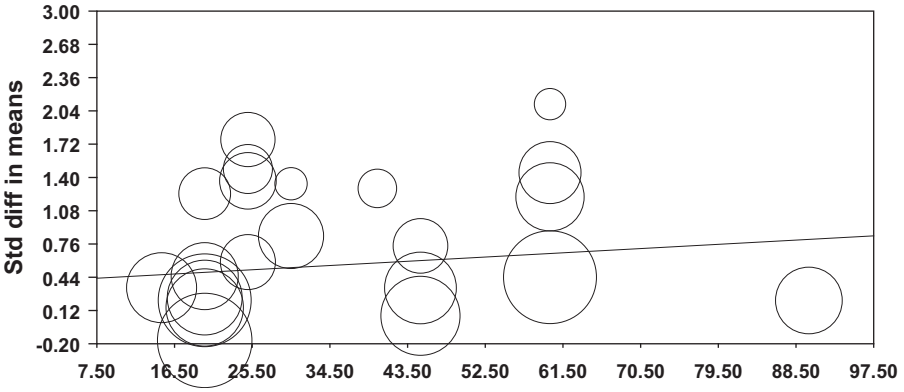


Figure 7. A meta-regression of the intensity of intervention.

information according to the cognitive characteristics of individuals with ASD. Furthermore, it can reduce the frequency and complexity of communication between individuals with ASD and intervention teachers (Sagayaraj et al., 2020). Besides, the TBI can reduce social withdrawal behaviors, anxiety, and fear in individuals with ASD (Kinsella et al., 2017). Many studies have shown that the TBI can improve the participation and pleasure of the ASD individuals, and increase the flexibility between intervention teachers and individuals with ASD (Artoni et al., 2018; Gal et al., 2016; Rosenbloom et al., 2019). The third possible reason is that TBI is easy to operate and can provide individuals with ASD the opportunity to acquire emotion recognition skills independently and effectively, thus ensuring positive intervention outcomes.

Notably, a study with wearable devices has a significant effect on the intervention of emotion recognition in individuals with ASD in recent years (Voss et al., 2019). Voss et al (2019) found that individuals with ASD who wear Superpower Glass (one modified of Google Glass) can rapidly "read" emotions from their parents by indicating emoji that appears in the Superpower Glass. Although this study results in positive influence, more researches are needed in the future.

For the other moderator variables, however, none of the categorical variables (country, parents' involvement, and the environment setting) moderated the overall effect sizes. It is worth noting that this study shows that the overall effect sizes were not moderated by the parent involvement. However, this conclusion should be interpreted with caution, as a large number of studies have shown that parent involvement in the emotion recognition intervention can improve the intervention effects. Therefore, parent's intervention is important and essential for emotion recognition. The reasons for the inconsistency may lie in the following two aspects: First, the TBI provides parents with fewer opportunities to participate in the intervention. The TBI presents dynamic personalized information, which reduces the frequency and complexity of communication

TABLE 5

A Meta-Regression of Continuous Variables on Emotion Recognition

Moderator variables	*Standard Error*	*LL*	*UL*	*Tau-squared*	*p*
Length of intervention	0.001	−0.021	0.045	0.22	0.46
Intensity of intervention	0.004	−0.003	0.012	0.21	0.22

among the ASD individuals which leads to improving the enthusiasm of participation. Secondly, parental characteristics (such as parents' age, anxiety level, and economic status) can influence the intervention effect of individuals with ASD, which means that the effectiveness of interventions without parent involvement is still influenced by the parental characteristics. However, due to insufficient data on the parent characteristics of ASD individuals in the literature of this meta-analysis, it is impossible to further discuss and analyze which of the parents-related factors influence the outcome of interventions.

For the continuous variables (the length of intervention period and the intensity of intervention) were not moderated the overall effect sizes. This conclusion supports the opinion of Sprung et al., (2015) who opined that the length of intervention period and the intensity of intervention were not related to the emotion recognition of ASD individuals.

Additionally, although it is important to investigate any moderating effects of age or gender distribution, nevertheless, the interpretation of meta-regression effects based on group average is problematic (Higgins & Thompson, 2002). Therefore, we recommend that future emotion recognition studies define specific age groups for their respective study sample (in addition to reporting the mean age, standard deviation, and range of the overall sample). This would allow for a better assessment of the influence of age on training effectiveness.

Limitations

This review on meta-analysis has some limitations. First, the number of included moderator variables is incomplete. We suggest that more moderator variables should be included in the study, such as post-test delay (days), peer tutoring, etc. which can be analyzed and discussed in more detail. Another limitation is that the number of recent studies is relatively small, especially the experimental studies on the TBI. The limited number of studies may affect the stability of the results. Finally, the included studies have language bias. In sum, the study shows that individuals with ASD may benefit from emotion recognition intervention and the results moderated the different intervention methods.

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